Machine Learning Algorithms for 5G and Internet-of-Thing (IoT) Networks

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***Abstract*— With the advent of 5G wireless networking, the Internet of Things (IoT) will be a network of sensors, actuators, and processing units for sensing, data sharing, and control. These are networks that have been specifically designed for a certain purpose. In general, the Internet of Things (IoT) offers information about all linked things. Rather than relying on a human operator, it performs all of the duties automatically. It is able to react immediately or via the experiences of the conditions. Ultra-low latency is one of the essential performance parameters for 5G to enable new delay-sensitive applications. To meet the 5G ultra-low latency goal, some changes to the network architecture are being considered. Different application needs necessitate different levels of transmission coverage, energy consumption, and communication technology. These factors all have an impact on performance and ultimately reduce the network lifetime of the huge IoT ecosystem.**

***Keywords*— IoT, 5G Network, Low Range Wide Area, Machine Learning.**

I. Introduction

The Internet of Things (IoT) allows billions of unconnected things to exchange data over a network without the need for a person to intervene. It establishes a connection between the physical world and the internet, allowing users to operate devices from a distance. From a person with a heart monitor implant to a car with sensors alerting the driver when the tyre pressure is low, the Internet of Things encompasses anything with an IP address, natural or artificial. One of the primary goals of the Internet of Things (IoT) is to make life easier for everyone by allowing everyday objects to communicate with one another and learn about our likes, dislikes, and wants in order to behave in a manner consistent with those preferences without the need for human intervention [1]. The Internet of Things (IoT) is changing our daily lives by tracking physical items in real time and allowing us to make better decisions. Sensor nodes in 5G networks are used primarily for two different objectives. It is critical for each sensor node to play a part in the collection of data from its immediate surroundings by employing a variety of sensors of different types. The "source function" is the term used to describe this. Nodes in close proximity to the sink will be required to process and transmit localised data to the sink via multi-hop distribution, which will take time [2]. The essential architecture of the Internet of Things is seen in Figure 1. A number of communication technologies, including Bluetooth, Zigbee, wireless local area networks, and 6LOWPAN, are used in these networks, all of which are powered by Bluetooth technology. Users can choose from a variety of various types of Personal Area Networks, depending on the platform they are using (PANs) [3]. PANs may communicate with one another and with other base stations through the internet. The sensor could be placed in a Bluetooth device that, after being detected and recorded by the sensor, communicates data to a base station via an internet connection after being transmitted to the base station.

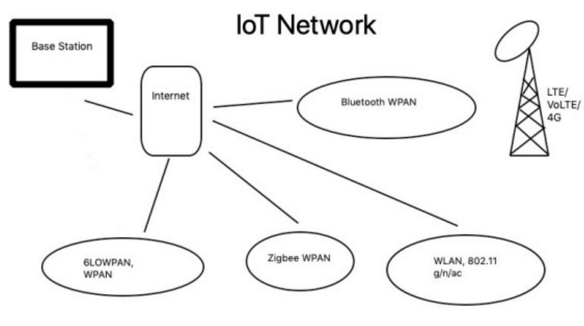

Fig. 1. IoT Architecture

Whether or not we need machine learning is a hot topic of debate. Supervised, unsupervised, and reinforcement-learning algorithms all fall under the umbrella term "machine learning." Real-time environments present a challenge to defining preferable data based on user needs [4-8]. When it comes to solving a real-world problem, machine learning techniques are designed to provide the best possible solution.

II. Background

A range of complex problems can be tackled by Machine Learning (ML) over the past several years. Facial recognition, social media services, and product suggestions are a few examples of possible applications. To be more specific, machine learning (ML) is concerned with creating algorithms and models that can learn from the data they are fed and utilise that information to make better judgments in the future (datasets) [9]. That manner, without any human help or input, computers may learn to adapt their activities. Supervised Learning (SL), Unsupervised Learning (USL), and Rule Extraction (RL) are the three main learning paradigms in ML, as well as four broad kinds of problems that can be performed. While SL employs labelled training datasets for classification and regression, USL uses unlabeled datasets to build models that can categorise sample sets into distinct groups, making it better suited to clustering challenges.

Network security, resource management, congestion control, and traffic classification and prediction and routing are just a few of the ML paradigms that have been applied to real-world networking issues [10]. Since its methods are simple to apply and because they may aid in decision-making tasks such as network scheduling, parameter adaptation, and resource allocation maximisation in dynamic contexts, RL has gained the most interest in recent years.

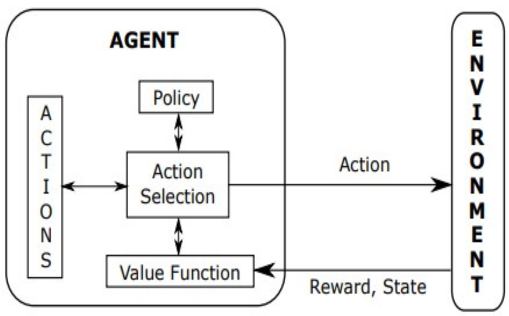

Fig. 2. Process of Renforcement Learning System

RL is a branch of machine learning that focuses on figuring out how to act in a way that maximises a long-term payoff. Learning takes place as a result of engaging with a real-world environment, and the focus is on achieving a certain objective. Instead of being told what to do, the learner is left to figure out which behaviours are most effective for him or her on their own [11]. In Fig. 2, the action options are selected based on value assessments as illustrated. To maximise long-term rewards, the agent focuses on activities that bring states of the highest value, rather than the highest reward. The estimation of values is a complicated procedure because they must be estimated and re-evaluated over the course of an agent's life. For the most part, the most critical part of every RL algorithm is some way to efficiently estimate values.

III. Methodology

From this perspective, LPWA networks use the star topology, shown in Figure 3, for the best performance. Only one powerful BS is needed at the centre of the start topology, whereas the EDs might be extremely simple. Unlike the ED, which wakes from sleep mode solely to transmit data, the BS is always listening for incoming messages. M2M technology's mesh structure was not its undoing, though. The fundamental issue is that there is no infrastructure in place. Cellular networks are designed from the start to cover a whole country [12]. However, traditional M2M networks often relied on a single base station (BS). For the vast majority of IoT/mMTC situations, mobility is essential. So the LPWA quickly superseded these technologies and became the fastest-growing IoT field. Sigfox currently covers more than 70 nations with its national coverage. Nothing from the past in the M2M space compares to this level of sophistication.

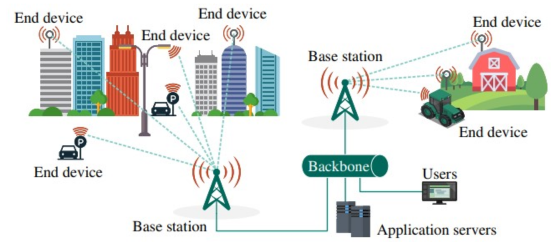

Fig. 3. Low power Wide Area Architecture

An adaptive modulation and coding (AMC) method, also known as link adaptation, explains the dynamic process of determining an optimal modulation and coding scheme that meets particular objectives (usually attaining an average BLER target or maximising throughput) under current channel conditions [13]. The downlink or uplink might be used for this procedure. Only the downlink is discussed here. If the downlink link adaption process relies on knowing the channel conditions at the user equipment (UE), then UE must continuously report a channel quality measurement to the base station (gNB). Base station channel quality can be utilised to figure out which MCS to employ for the next downstream transmission.

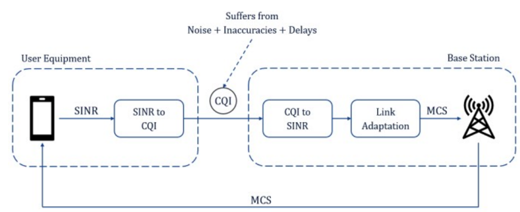

Fig. 4. Link Adaption process for 5G IoT Network

Sensors and cameras are used in IoT devices to obtain data about their surroundings. All the acquired information is often transmitted in pockets, which means that it is split up into a variety of transmissions [14]. The QTTP queues a large number of nodes before accumulating them in the server, where they can be retrieved for later use.

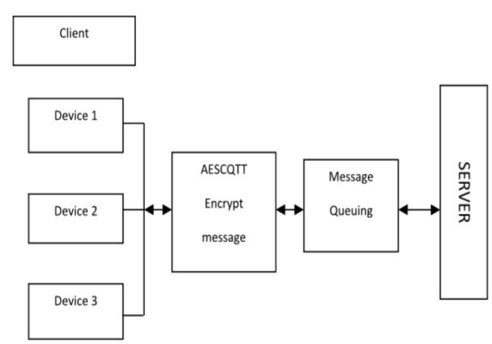

Fig. 5. Proposed work Architecture

Data packets travelling across the collected data are encrypted using a 192-bit key and then saved in the server if node or device-1 is given the key size of 192 bits. The suggested randomization approach also insured that no consecutive nodes would have the same key size for encryption [15]. Key size is assigned to the node when data transmission is complete. It can be 128 or 256 bits. Key size information is updated at the confirmed user end, so the data is protected during transmission.

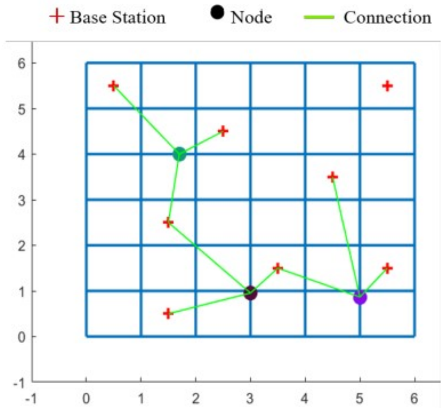

Fig. 6. RSS based Proposed system Model

As depicted in Figure 6, a 6-by-6-kilometer road network with 1 km-distance intersections is created. Eight BSs are randomly placed in this grid. Randomly created on a road, a mobile node has a starting speed of 8-12 km/h for pedestrians and 55-65 km/h for cars, selected from a uniform distribution. A mobile node's speed is supposed to remain constant until it leaves the road network. At any given intersection, a mobile node has a 50% probability of moving forward and a 25% chance of turning right or left. The RSS values of the three BSs closest to a node in the road network are measured. For the purpose of demonstrating RNNs' capacity to learn and anticipate RSS values in advance, this approach was selected.

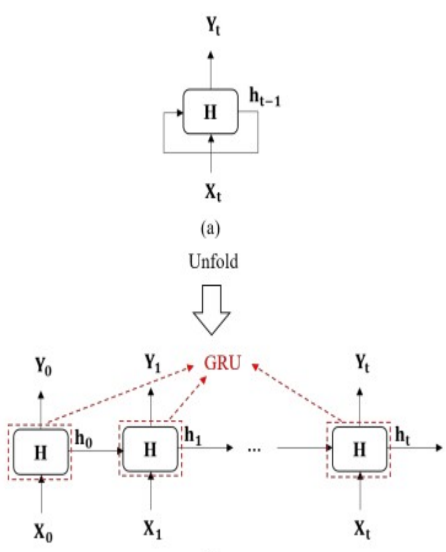

Fig. 7. Structure of RNN

Eight-element feature vectors are utilised to represent BSs and their RSS values, and the positions correlate to the BSs' RSS values. BSs within 500 ms of the mobile node record RSS values every 500 ms. There are no values for the other vector elements. When the mobile node moves through the road network, the RSS values (feature vectors) are collected and stored sequentially in a queue data structure. Whenever the queue size reaches 150 RSS pieces, the oldest ones are removed from the queue and no longer considered part of the queue's content. This approach was used to simulate mobile nodes up to 100,000 times.

IV. Results

The best trigger conditions for CoMP activation and VC creation were predicted utilising 512 Gated Recurrent Units (GRUs) in the Gated Recurrent Unit (GRU) mechanism in simulations employing RSS values for proactive judgments on whether or not to build or dissolve VCs. Training and testing the GRU model required the utilisation of almost 70,000 sequences. The testing error substantially decreased with 15 training epochs and 75 steps per epoch of training. The GRU model was able to accurately predict the RSS values of two unique UEs by using the three closest BSs. Most of the time, the GRU model's predictions for RSS values are right on the money.. The network can take proactive action based on these projections and enable CoMP transmission in order to provide a consistent user experience.

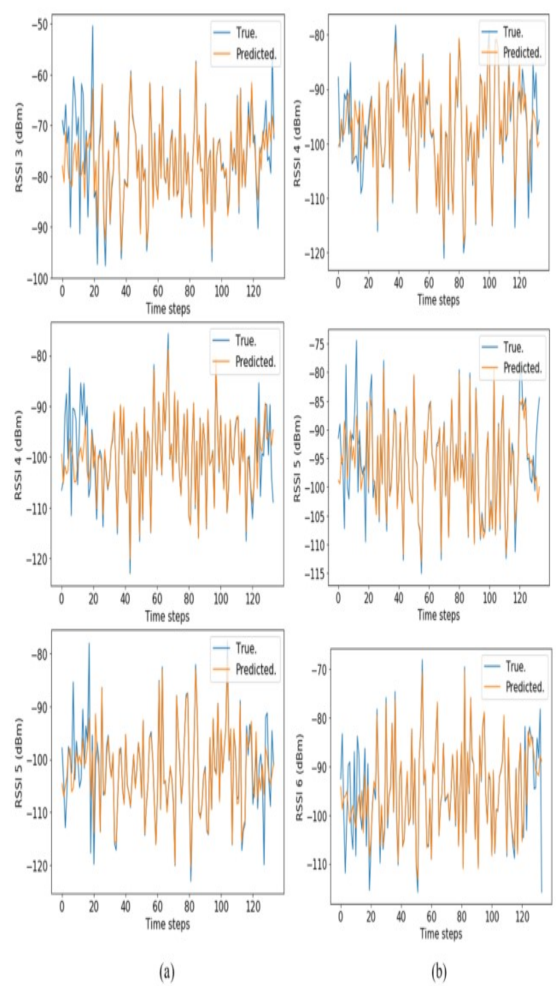

Fig. 8. Plot for comparision of true and predicted signal strength

As shown in Figure 9, when a GRU-RNN predictive model is used, the CDF of the number of virtual cells that are enabled is shown. Instead of relying on a static virtual cell, the virtual-cell mode is activated 14 times over the course of the nodes' time in the network, with a chance of around 0.95. Figure 10 shows how the loss function gradually decreases as the number of epochs increase. MSE testing error gradually decreases after 75 training epochs.

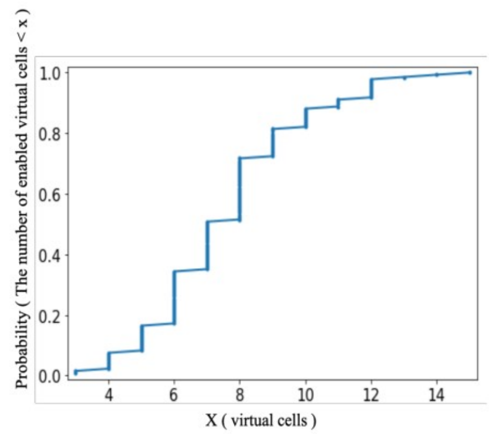

Fig. 9. Using the RL predictive model to enable virtual cells.

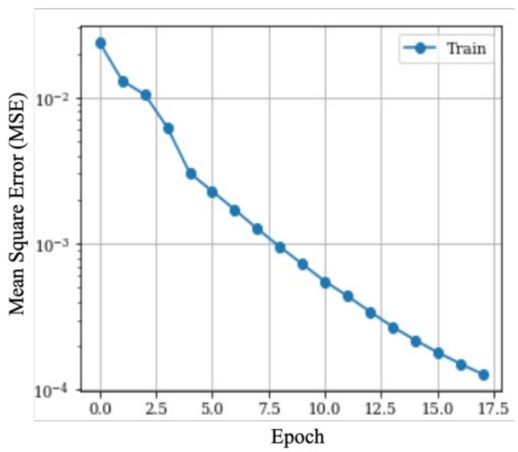

Fig. 10. Convergence of the MSE loss function training model.

V. Conclusion

The fundamental goal of this research is to develop new MAC layer protocols that are simple to design, energy efficient, scalable, and compatible with the low-complexity requirements of Internet of Things devices. Predictive mobility management in 5G wireless networks was tested using a deep machine learning (ML) technique. CoMP transmission trigger circumstances and VC selection were investigated using ML. To maintain a consistent user experience, VCs are generated dynamically in the suggested method. The suggested algorithm also enables the mobility management function to promote VC formations due to the low-latency needs of next-generation 5G networks.

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